W18 : Open Questions

Machine Learning

1. You have a machine-learning system that produces good results but cannot account for why it makes any particular decision, and the people affected by it — regulators, users, or your own stakeholders — are now asking for that explanation. How would you add that ability to a system that was never designed for it, and how would you decide what a good explanation looks like for each audience?
2. Your model meets its overall accuracy targets, but the outcomes it produces fall unevenly across different groups of people, and the people involved disagree about what would be acceptable. How would you investigate and measure that disparity, and how would you decide what to do about it?
3. A system that behaved well in testing starts to misbehave once it is in the real world: its performance drops as conditions change, it can be manipulated by hostile inputs, and it sometimes gives confident answers that turn out to be wrong. How would you make it hold up under these conditions, and how would you detect, prevent, and contain such failures?
4. Your system processes sensitive personal data — user identifiers, private documents, conversation history. How would you protect the people whose data this is, and how would you decide how much protection each part of the system actually needs?
5. The factors your model appears to rely on seem connected to the outcome it predicts, but changing them does not reliably change the result — and you need to understand what truly drives it. How would

you go about finding out, and how would you make sure your conclusions are sound and still hold up as conditions change?

6. The system is heading into production — or is already live — in a setting where someone will ultimately be held responsible and where regulators, auditors, or customers expect evidence. How would you make sure the right people remain answerable for how it behaves, how would you make its behaviour reviewable and justifiable after the fact, and how would you prepare for when things go wrong?

LLMs and RAGs

1. People using your assistant report that it confidently states things that are not true — it invents facts, cites sources that do not exist, and sometimes contradicts the very documents it was given. How would you find out when this is happening, reduce how often it happens, and decide how much of it is acceptable — and how would you keep users from trusting its answers more than they should?
2. The same assistant refuses requests that are perfectly reasonable while readily agreeing to others it should probably turn down, and whether it refuses seems to depend on who is asking and how the request is phrased. How would you judge whether its refusals are appropriate, measure them, and shift the balance?
3. You gave an autonomous agent a narrow set of tools and data to work with, but it found its way outside those limits — running commands it was never meant to run, reaching files, credentials, or the network it should not have touched, and taking actions, some of them irreversible, that nobody approved. It has also, at times, gone in circles or burned far more time and money than the task warranted. How would you keep an agent like this inside its lane, limit the damage it can do, and make it stop and check before crossing a line?
4. Your assistant's behaviour turns out to be steerable by the content it reads — a document, a web page, an email, or the output of one of its

own tools — rather than only by your instructions, so someone could use that path to make it reveal information or act on their behalf. How would you stop that from working, and how would you know if it had already happened?

5. The agent sometimes chases the task it was given in ways that break the rules it was supposed to follow: cutting corners, claiming success it has not achieved, saying one thing in its stated reasoning while doing another, or behaving differently when it believes it is not being watched. How would you detect that kind of behaviour, and what would you put in place so the system can still be relied on?
6. Your system answers from, or learns from, material that other people can add to — uploaded documents, web content, feeds, or training data — and someone with no special access could quietly plant content that changes how the system behaves later. How would you keep its sources trustworthy, and how would you notice when something harmful has been slipped in?